

Energy System Modelling Using OSeMOSYS

Hands-on 5

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OSeMOSYS UI software:

Climate Compatible Growth. (2024). MUIO (Version v5.0.0). GitHub. <https://github.com/OSeMOSYS/MUIO/releases>

Learning outcomes

By the end of this exercise, you will be able to:

- Define existing thermal power plants taking in fuel to generate electricity
- Define the existing transmission network
- Define the existing distribution network
- Run the model and check results on production by technology and installed capacity of each technology

Define existing thermal power plants taking in fuel to generate electricity

In Lecture 6, we learnt how to represent a technology in OSeMOSYS, and which parameters characterize thermal power plants and transmission and distribution technologies.

In this Hands-on, we will add 4 technologies in total: 2 thermal power plants, 1 technology representing the transmission system and 1 for the distribution network. Two new fuels will be added to the model: ELC001 (Electricity coming directly from power plants) and ELC002 (Electricity after transmission).

In order to represent a thermal power plant, remember that the following parameters must be considered:

- InputActivityRatio: defines the rate of fuel consumed (e.g., diesel)
- OutputActivityRatio: defines the fuel provided (i.e., Electricity)
- CapacityToActivityUnit: used to convert data related to the Capacity of technology into the Activity it can generate (for power generation, e.g., PWRDSL, this value should be set to 31.536).
- Fixed Cost: defines the fixed Operation & Maintenance cost (\$/kW)

- CapitalCost: defines the overnight investment cost of the plant (\$/kW)
- OperationalLife: defines the lifetime of the technology (in years)
- ResidualCapacity: defines the existing capacity of the technology (in GW) and its expected decommissioning.
- Capacity Factors: represents the variability in generation at each point in time.

Try it: Let's add PWRDSL - the technology representing a diesel power plant.

Now let's add 'ELC001' (Description: Electricity from Power plants) as a commodity. Then go to the technologies tab of the model configuration page and add 'PWRDSL'. Keep the unit of capacity as GW and the unit of activity as PJ. Add the data for PWRDSL as given in the Data Preparation File OSeHO5. Pay close attention to the CapacityFactor values. Follow the previous steps when you add PWRNGS using the data provided in the Data Preparation File OSeHO5.

Define the existing transmission network

We will repeat the exercise once more giving the example of a technology which represents the transmission network (PWRTRN). When representing the transmission technology, the following parameters must be considered:

- InputActivityRatio: defines the rate of fuel consumed (i.e., Electricity from power plants)
- OutputActivityRatio: defines the fuel provided (i.e., Electricity)
- CapacityToActivityUnit: 31.536
- Fixed Cost, CapitalCost, OperationalLife, ResidualCapacity.

Try it: Let's add PWRTRN - the technology representing the transmission grid. Add a new commodity which is 'ELC002'. Add the technology 'PWRTRN' with the input being ELC001 and output ELC002. Add the data for PWRTRN as given in the Data Preparation File OSeHO5.

Define the existing distribution network

We will repeat the exercise once more giving the example of a technology which represents the distribution network (PWRDIST). Let's add PWRDST - the technology representing the distribution network. Set the input as ELC002 and output as ELC003. Add the data for PWRDST as given in the Data Preparation File OSeHO5.

Run the model and check results

You should obtain the results by running the model in the user interface as learnt in the hands-on 3. Notice that the production of PWRDIST is equivalent to the electricity demand, as this technology delivers the final electricity to the system. Conversely, you can verify that the electricity production of ELC002 by PWRTRN equals the energy demand multiplied by the input activity ratio of PWRDIST.